



CIVIL MAINT. Department
SIMHADRI

Note No :1

Ref:SMTTP/Civil Maint/2024-25/LMI/316965

Date:

16/07/2024(17:39:44)

Subject:Review of LMI Manual for inspection maintenance of machine foundations including spring supported foundations with Ref. LMIOGNOPSCIV011

Sub : Review of LMI Manual for inspection & maintenance of machine foundations including spring supported foundations with Ref.LMI/OGN/OPS/CIV/011

The LMI of "Manual for inspection & maintenance of machine foundations including spring supported foundations" with Ref. LMI/OGN/OPS/CIV/011, Rev. No.5 was prepared and approved in the month of Feb'2022.

The LMI has been reviewed as on date and no changes are envisaged.

Put up for approval please.

Submitted by:

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Note No:3

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Note No:4

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN. Checklist May be added

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Note No:5

LMI is prepared as per standard format of Simhadri LMI. Checklist added. Revised LMI attached.

The list of Modifications in the LMI revision are as under:

1. Following is added at the end of Clause 7 (II) in-line with revision (a) in OGN Rev.01 May 2024:

“Station must have/procure “Demountable mechanical” range of mechanical strain measurement devices to assess in-situ Strain measurement.”

2. Revision (b) in OGN Rev.01 May 2024 regarding TG benchmark reading of Supercritical Units is **not added** as the same is applicable for 660/800MW units and not for 500MW Units at Simhadri.

3. Clause 9: Responsibility of carrying out vibration analysis has been changed from **DGM (MTP-Condition monitoring)** to **MTP-Condition monitoring**.

4. Checklist, Appendix-I (Vibration Troubleshooting Chart) and Appendix - II (Procedure for In-situ Mechanical Strain Measurement) added as per OGN.

LMI is submitted for Approval please.

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Please review.

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Note No:8

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Note No:9

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LMI reviewed and no changes proposed.

Forwarded for approval.

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Note No:16

Forwarded for further approval

आगे की स्वीकृति के लिए भेजा गया

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Note No:17

Approved as HOP(Simhadri) is out of station

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**Manual for Inspection and Maintenance of Machine Foundations
Including Spring Supported Foundations**

**LMI/OGN/OPS/CIV/011
REVISION NO.:06; July'2024**

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Manual for Inspection and Maintenance of Machine Foundations Including Spring Supported Foundations
2	LMI No.	LMI/OGN/OPS/CIV/011 Rev:06
3	Reference Document	COS-ISO-00-OGN-OPS-MECH-055 Rev. No.: 01 May 2024
4	Superseded Document	LMI/OGN/OPS/CIV/011 ISSUE NO.1; REVISION NO.: 05; FEB'2022
5	Prepared By	Name: Chandrasekhar Godi, DGM (O&M- Civil)
6	Checked By	Name: M C Yadav, DGM (O&M- Civil) Name: Mahesh Alugolu, AGM (O&M-Civil) Name: Ajay Gupta, DGM (EEMG) Name: Rajesh Chawla, AGM (MTP) Name: Om Prakash, AGM (BMD) Name: Navin Kishore, AGM (TMD) Name: A R Koteswara Rao, AGM (FM) Name: D K Tomar, AGM (TS)
7	Reviewed By	Name: S.J.David GM (O&M)
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TITLE	Manual for Inspection and Maintenance of Machine Foundations Including Spring Supported Foundations
LMI No.	LMI/OGN/OPS/CIV/011 Rev:06

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TITLE	Manual for Inspection and Maintenance of Machine Foundations Including Spring Supported Foundations
LMI No.	LMI/OGN/OPS/CIV/011 Rev:06

1.0 INTRODUCTION:

Machine foundations at simhadri are engineered and erected to withstand and safely transmit to the ground all static and dynamic forces exerted by the machines. Essentially the machine foundations are classified as

- a) Block foundations in RCC
- b) Framed foundations in RCC or structural steel
- c) Spring supported foundations

All block foundations supporting rotating equipment resting on soil or piles are designed using elastic half space theory. The mass of RCC block has been maintained greater than three times the mass of machine. The foundations of the turbo generator, boiler feed pump, fans, mills, and coal crusher are engineered with detailed static and dynamic analysis and are supported on vibrations isolated system. The vibration isolated system consists of helical spring units and viscous dampers, supporting the RCC deck supporting the machine. While turbo generator, BFP, and fan foundations are designed with isolation efficiencies greater than 90% mill foundations are designed with isolation efficiencies greater than 70%. The normal spring capacity in these foundations (except in mills) is at least 25% greater than the actual supported weights and is 35% higher for mills.

A good static and dynamic performance of the machine foundation is vital for the smooth operation of the machine supported by it. Hence, it is, necessary to keep a watch on the machine foundation to know about any impending problems, as dynamic forces being time varying in nature can cause reversal of stresses in structure, which in long run can reversal of stress in the structures, which in long run can cause fatigue.

The objective of this LMI is to instruct & monitor the measures to be taken for inspection of machine foundations.

2.0 SCOPE:

The scope of this LMI is to detail the methods that are to be adopted for the inspection of machine foundations.

3.0 REFERENCE DOCUMENT:

Operating guidance note NO. OGN/OPS/CIV/011 circulated by operation services NTPC corporate center.

4.0 SUPERSEDED DOCUMENT:

Inspection of Machine Foundations REV.05 DATED Feb' 2022

5.0 Details of Revision

TITLE	Manual for Inspection and Maintenance of Machine Foundations Including Spring Supported Foundations
LMI No.	LMI/OGN/OPS/CIV/011 Rev:06

Following is added at the end of Clause 7 (II) in-line with revision (a) in OGN Rev.01 May 2024:

“Station must have/procure “Demountable mechanical” range of mechanical strain measurement devices to assess in-situ Strain measurement.”

Revision (b) in OGN Rev.01 May 2024 regarding TG benchmark reading of Supercritical Units is **not added** as the same is applicable for 660/800MW units and not for 500MW Units at Simhadri.

Clause 9: Responsibility of carrying out vibration analysis has been changed from **DGM (MTP-Condition monitoring)** to **MTP-Condition monitoring**.

Checklist, Appendix-I (Vibration Troubleshooting Chart) and Appendix - II (Procedure for In-situ Mechanical Strain Measurement) added as per OGN

6.0 SPECIFIC INSTRUCTIONS:

The following aspects are to be focused to carry out effective inspection of foundations (OS/PE) for further study and rectification.

- FFT (fast fourier transformation) analysis (signature analysis) of vibration data machine supported by foundations.
- Visual inspection and measurement of strain in concrete.
- Survey of steel structures supporting machines.
- Inspection of spring damper system

7.0. METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS:

I. VIBRATION ANALYSIS:

This is not applicable for foundations supporting minor rotating equipment weighing less than one tonne or if the mass rotating parts or less than hundredth of mass of foundation:

- Check vibration levels at the bearings in vertical, horizontal and axial directions, once in a month **MTP**
- Whenever vibration limit as measured at Sl. No. A are exceeding the limits, as detailed below for major equipment (as specified below) or as given O&M manuals carry out FFT analysis to diagnose the cause and report the same.....**MTP**

LIMITS:

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MAJOR EQUIPMENT	AMPLITUDE (RMS AT 50HZ) (DISPLACEMENT/VELOCITY) MICRONS: MM/SEC
Turbo generator	25-40/6-8
ID,PA,FD fans	-----/5-10
Coal crushers	-----/14-16
Boiler feed pumps	-----/7-10
Mills	-----/5-7
Seawater P/Pas	-----/9-10
CW P/p's	75-95/5-6

If the cause as analysed at SI.NO.B can traced to foundation, report the same to project engineering & operation services. -----MTP

II. VISUAL INSPECTION AND MEASUREMENT OF STRAIN IN CONCRETE:

These activities are to be carried out on yearly basis

- A) Visual inspection of concrete structure are to be carried out for any cracks, spillings etc. (Falling over of cover concrete) ----**Inspection committee.**
- B) In case of cracks mapping of cracks to be carried out ---- **Inspection committee.**
- C) Checks for rusting in reinforcement steel. ----- **Inspection committee.**

Station must have/procure Demountable mechanical range of mechanical strain measurement devices to assess in-situ Strain measurement.

III. VISUAL INSPECTION OF STEEL STRUCTURES SUPPORTING MACHINES:

These activities are to be carried out on yearly basis

- A) Checking the structural for corrosion -----**Inspection committee**
- B) Checking top welding joints for cracks ----- **Inspection committee**
- C) Checking the structures for local vibrations -----**Inspection committee**

IV. INSPECTION OF SPRINGS-DAMPER SYSTEM:

These activities are to be carried out on yearly basis

- A) Checking of top and bottom plates for corrosion ----- **Inspection committee**
- B) Checking of spring coils for corrosion -----**Inspection committee**
- C) Checking springs coils for breakage -----**Inspection committee**

V. LIST OF MACHINE FOUNDATIONS TO INSPECTED BY THE COMMITTEE:

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(FOR STAGE 1&2)

- 1) TG
- 2) IDFANS
- 3) PA FANS
- 4) FD FANS
- 5) BFP'S
- 6) COAL CRUSHERS
- 7) COAL MILLS
- 8) SEA WATER PUMPS
- 9) CW PUMPS
- 10) IAC AND PACs
- 11) ASH SLURRY PUMPS

8.0 EXCEPTION REPORTING

The exceptions encountered during inspection are to be reported as per the following format ----- **Inspection committee**

Description of Work	Unit No	Location of exception	Remarks

9.0 RESPONSIBILITY

The responsibility of carrying out different tasks in relation to inspection of machine foundation in plant areas as detailed in the LMI are with the inspection committee consisting of representatives of O&M-Civil, CHP, MM (TMD & BMD) & FQA. This shall be followed as earlier office order Ref. No. SMPP/GM(O&M)/2021, dtd: 19.03.2021 and members of the committee shall be changed if required. The responsibility of carrying out vibration analysis is with **MTP-Condition monitoring**.

10.0 RECORD KEEPING

The records to be maintained under the LMI Include

- a) Compliance record of the activate detailed under this LMI
- b) Reports of inspection carried out
- c) Exception in reporting

The above records are to be maintained by MTP, MMD & O&M-Civil for relevant responsibilities.

11.0 REVIEW

HOD (Civil) will review & take the approval of Competent Authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required.

12.0 Checklist:

Checklist for Inspection of Machine Foundations

(A) Concrete Structures				
S.No	Description	Responsibility	(Yes/ No)	Remark

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1.	The structures should be inspected every six months for any cracks, spalling, stains etc.	Inspection committee		
2.	In case cracks are noticed, their pattern should be mapped.	Inspection committee		
3.	If cracks or stains are noticed, cover concrete should be chipped off at some of the crack locations to check the reinforcement for corrosion.	Inspection committee		
4.	Increase in crack width should be monitored by means of a glass strip or mortar strip fixed across the cracks. Precise measurement can also be done by 'Demountable mechanical strain gauge.	Inspection committee		
5.	In case of spalling, the reinforcement should be checked for corrosion. The structure will have to be checked for structural inadequacy / distress.	Inspection committee		
6.	Likely presence of voids in the structure can be monitored once in a year at suspected areas only by using ultra sonic velocity test.	Inspection committee		

(B) Steel Structures

1.	Check for corrosion of members	Inspection committee		
2.	Check weld joints for any incipient cracks	Inspection committee		
3.	Check local vibrations of members	Inspection committee		
4.	Check for source of moisture that can lead to oxidation and eliminate for	Inspection committee		

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	improved life and performance.			
(C) Spring Supported Foundations				
1.	Check spring heights (as per inspection schedule recommended by OEM) and compare with Original/Erection protocol	Inspection committee		
2.	Check top and bottom plates for corrosion	Inspection committee		
3.	Check spring coils for corrosion	Inspection committee		
4.	Check spring coils for breakage	Inspection committee		
5.	Check viscous dampers for water tightness	Inspection committee		
6.	Measurement of TG deck benchmark readings by laser method for newly commissioned 660/800MW Super critical Units. (Six monthly frequency)	TMD		Not Applicable for Simhadri
(D) Vibration Problems				
1.	Check vibration levels at the bearings in vertical, horizontal and axial directions.	MTP-Condition monitoring		
2.	Whenever vibration levels are exceeding the limits, carry out vibration analysis to diagnose the cause and carry out rectification measures.	MTP-Condition monitoring		
3.	More often than not, the source of high vibrations lies in the machine. It	Respective Maint.		

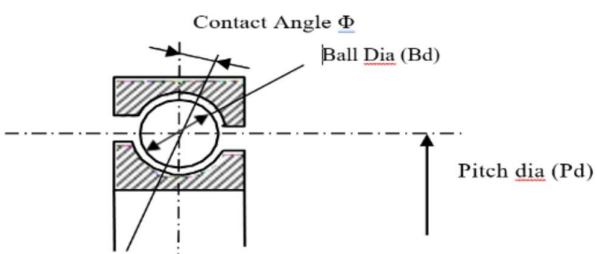
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	should be suitably rectified.			
4.	In case the problem is traced to the foundation, the data should be passed on to OS / Project Engineering for a suitable solution.	Civil (O&M)/ Respective Maint.		

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APPENDIX – I Vibration Troubleshooting Chart

Nature of fault	Frequency of Dominant Vibration (Hz=rpm/90)	Direction	Remarks
Rotating Members out of Balance	1 x rpm	Radial	A common cause of excess vibration in machinery
Misalignment & Bent Shaft	Usually 1 x rpm. Often 2 x rpm Sometimes 3 & 4 x rpm	Radial & Axial	A Common fault
Damaged Rolling Element Bearings (Ball, Roller etc.)	Impact rates for the individual bearing component Also vibrations of high frequencies (2 to 60 kHz) often related to radial resonance's in bearings	Radial & Axial	Uneven vibration levels often with shocks. Impact Rates (Hz) For Outer Race Defect (Hz)=$n/2L(1-Pd/Bd\cos\Phi)$ For Inner Race Defect (Hz)=$n/2L(1+Pd/Bd\cos\Phi)$ For Ball Defect (Hz) =$Pd/BdL[1-(Pd/Bd\cos\Phi)]$  N= Number of Balls or Rollers L= Rotative rev./ sec. Between inner & outer races
Journal bearings Loose in Housing	Sub-harmonics of shaft rpm exactly 1/2 or 1/3 x rpm	Primarily Radial	Looseness may only develop at operating speed and temperature (e.g. turbo-machines).
Oil fires Whirl or whip in Journal Bearings	Slightly less than half shaft speed (42% to 48%)	Primarily Radial	Applicable to high-speed (e.g. turbo) machines
Hysteresis Whirl	Shaft critical speed	Primarily Radial	Vibrations excited when passing through critical shaft speed are maintained at higher shaft speeds. Can sometimes be cured by checking tightness of rotor components.
Damaged or Worn gears	Tooth meshing	Radial & Axial	Sidebands around tooth meshing frequencies indicate modulation (e.g. eccentricity) at

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	frequencies (shaft rpm x number of teeth) and harmonics		frequency corresponding to sideband spacings. Normally only detectable with very narrow-band analysis and cepstrum
Mechanical Looseness	2 x rpm		Also sub-and inter-harmonics as for loose journal bearings
Faulty Belt Drive	1,2,3 & 4 x rpm of belt	Radial	The precise problem can usually be identified visually with the help of stroboscope
Unbalanced Reciprocating Forces and couples	1 x rpm and / or multiples for higher order unbalance	Primarily Radial	
Increased Turbulence	Blade & Vane passing frequencies and harmonics	Radial & Axial	Increasing levels indicate increasing turbulence
Electrically induced vibrations	1 x rpm or 1 or 2 times synchronous frequency.	Radial & Axial	Should disappear when turning off the power

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Appendix - II Procedure for In-situ Mechanical Strain Measurement

The Demountable mechanical range of mechanical strain measurement devices are ideally suited for the measurement of strain in concrete structures. The instrument consists of an Invar main beam with two conical locating points, one in a fixed position at one end, and the other pivoting on a special knife edge. This pivoting movement is transmitted to a dial gauge graduated in 0.002 mm mounted on a table. The main beam is fixed to the pivot block and dial table assembly, and the thermal movement between the beam and the table takes place along a grooved seating. The expansion of the plunger of the dial gauge is thus closely counter-balanced by the table expansion, and the gauge requires only very small temperature corrections during use. A standard Invar beam is supplied for this. Location of the gauge points is made by drilled stainless steel discs cemented to the structure, using a setting-out bar.